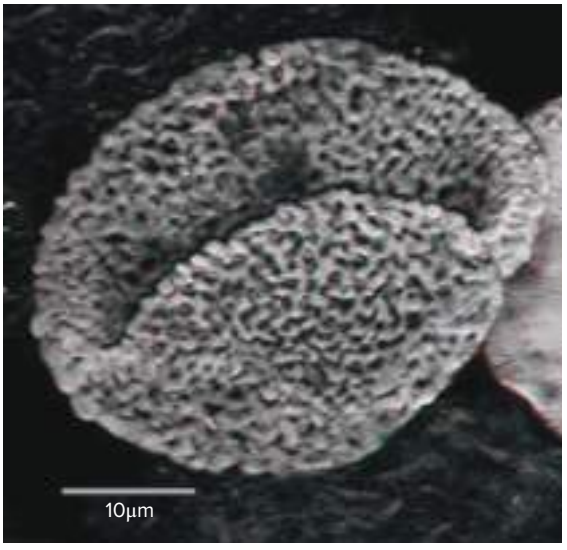
Nanjinganthus dendrostyla, dating back 174 million years to the early Jurassic. *N. dendrostyla* is a stunning fossil with seeds and flowers, giving a glimpse into what some of the early flowering plants may have looked like (see Fig. 19.8). Unlike the soft parts of plants, pollen does preserve well, has characteristic markings, and can be used to date the appearance of a seed plant in an area. Paleobotanists working with drilling cores from Switzerland discovered 240-million-year-old fossilized pollen from the Triassic, pushing the origin of flowering plants back even further (Fig. 23.20*a*).

To find the angiosperm of today that might be most closely related to the first angiosperms, botanists have turned to DNA comparisons. Gene-sequencing data singled out *Amborella trichopoda* (Fig. 23.20*b*) as having ancestral traits. This small, woody shrub, with small, cream-colored flowers, lives only on the island of New Caledonia in the South Pacific. Its flowers are about 4–8 mm wide, and the petals and sepals look the same; therefore, they are called tepals. Plants bear either male or female flowers, with a variable number of stamens or carpels.



a. Fossil pollen


b. *Amborella trichopoda*

Figure 23.20 Evolution of flowering plants. **a.** Flowering plant pollen from the Triassic. **b.** Analysis of molecular data suggests that the plant seen here, *Amborella trichopoda,* is the most closely related to the first flowering plants. (a): Susanne Feist-Burkhardt; (b): Doukdouk/Alamy Stock Photo

Although *A. trichopoda* may not be the original angiosperm species, it is sufficiently close that much may be learned from studying its reproductive biology. Botanists hope that this knowledge will help them understand the early adaptive radiation of angiosperms during the Tertiary period. The gymnosperms were abundant during the Mesozoic era but declined during the mass extinction that occurred at the end of the Cretaceous period (145.5–65.5 MYA). Angiosperms survived and went on to become the dominant plants during modern times.

Monocots and Eudicots

Most flowering plants belong to one of two classes. These classes are the *Monocotyledones,* often shortened to simply the **monocots,** and the *Eudicotyledones,* shortened to **eudicots.** The term *eudicot* (meaning "true dicot") is more specific than the term *dicot.* It was discovered that some of the plants formerly classified as dicots diverged before the evolutionary split that gave rise to the two major classes of angiosperms. These earlier-evolving plants are not included in the designation eudicots.

Cotyledons (Gk. *kotyledon,* "cuplike cavity") are the "seed leaves" containing the nutrients that nourish the plant embryo. If a seed has one cotyledon, it is a monocot, and if a seed has two cotyledons, it is a eudicot (see Fig. 27.8). Several common monocots are corn, tulips, pineapples, and sugarcane; common eudicots include cacti, strawberries, dandelions, poplars, and beans. Table 23.2 lists several fundamental features of monocots and eudicots.

The Flower

Although **flowers** vary widely in appearance (Fig. 23.21), most have certain structures in common (Fig. 23.22).

- The **sepals,** collectively called the calyx, protect the flower bud before it opens. The sepals may drop off or may be colored like the petals. Usually, however, sepals are green and remain attached to the flower stalk.
- The **petals,** collectively called the corolla, are quite diverse in size, shape, and color. The petals are often used to attract a particular pollinator.
- Next are the **stamens.** Each stamen consists of two parts: first, a slender stalk, called a filament, that holds up a second

Table 23.2 Key Features of Monocots and Eudicots

Monocots	Eudicots
One cotyledon	Two cotyledons
Flower parts in threes or multiples of three	Flower parts in fours or fives or multiples of four or five
Pollen grain with one pore	Pollen grain with three pores
Usually herbaceous	Woody or herbaceous
Usually parallel venation	Usually net venation
Scattered bundles in stem	Vascular bundles in a ring
Fibrous root system	Taproot system

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a. Monocot



b. Eudicot

Figure 23.21 Flower diversity. Regardless of size and shape, flowers, such as this (**a**) monocot and (**b**) eudicot, share certain features.
(a): Ed Reschke/Stone/Getty Images; (b): Steven P. Lynch

structure, a saclike container called the anther. Pollen grains develop from microspores produced within the anther.

- At the very center of a flower is the **carpel,** a vaselike structure with three major regions: the stigma, an enlarged, sticky knob; the style, a slender stalk; and the **ovary,** an enlarged base that encloses one or more ovules. The ovule becomes the seed, and the ovary becomes the fruit. Fruit is often instrumental in the distribution of seeds.

Note that not all flowers have all these parts (Table 23.3). A flower is said to be *complete* if it has all four parts; otherwise, it is *incomplete.*

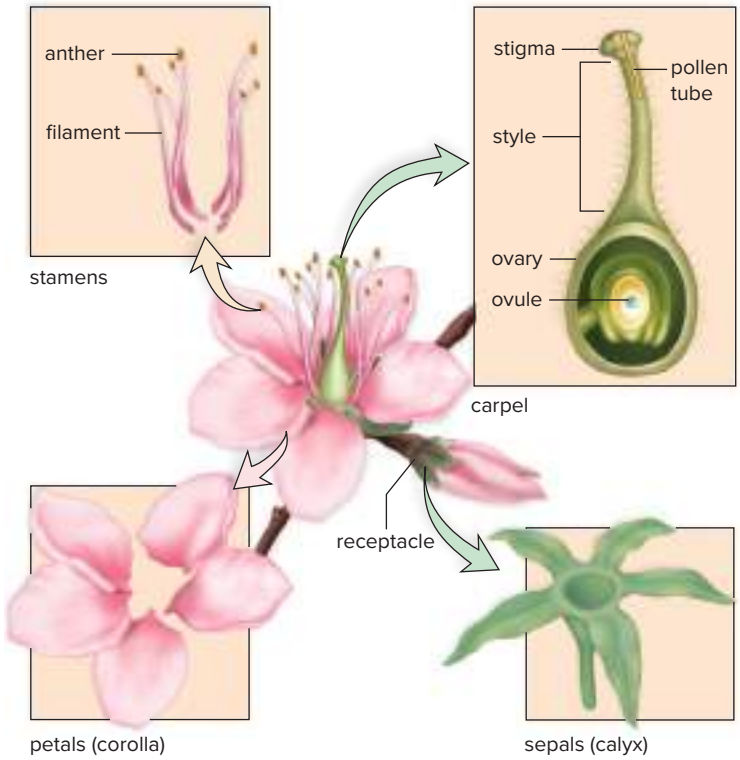


Figure 23.22 Generalized flower. A flower has four main parts: sepals, petals, stamens, and carpels. A stamen has an anther and filament. A carpel has a stigma, style, and ovary. An ovary contains ovules.

Table 23.3 Other Flower Terminology

Term	Description of Flower
Complete	All four parts (sepals, petals, stamens, and carpels) present
Incomplete	Lacks one or more of the four parts
Perfect	Has both stamens and (a) carpel(s)
Imperfect	Has stamens or (a) carpel(s), but not both
Inflorescence	A cluster of flowers
Composite	Appears to be a single flower but consists of a group of tiny flowers

Flowering Plant Life Cycle

Figure 23.23 depicts the life cycle of a typical flowering plant. Like the gymnosperms, flowering plants are heterosporous, producing two types of spores. A megaspore located in an ovule within an ovary of a carpel develops into an egg-bearing female gametophyte, called the embryo sac. In most angiosperms, the embryo sac has seven cells; one of these is an egg, and another contains two polar nuclei. They are called the polar nuclei because they came from opposite ends of the embryo sac.

Microspores, produced within anthers, become pollen grains that, when mature, are male gametophytes with sperm. The mature male gametophyte consists of only three cells: the tube cell and two sperm cells.

During pollination, pollen is transported from the anther to the stigma of a carpel. Here, the tube cell produces a pollen tube that

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